The Mathematics Underlying Elliptic Curve Cryptography

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University of Warwick

Contents

1 Introduction

We split this paper into two parts:
The first part introduces elliptic curves in an extension of Euclidean space, namely projective space and we establish the notion of the *point at infinity* in this space. This will allow us to illuminate our first key theorem, the group structure present in the set of points on a given elliptic curve union the point at infinity. Part I continues to discuss finite fields and we prove our second key theorem which states every finite field is isomorphic to $\mathbb{F}_{p^k}$, which is the finite field with p^k elements for some prime p and positive integer k.
In Part II, we introduce what a cryptosystem is, how it works and this motivates the *discrete logarithm problem* which is the heart of any cryptographic system. In short, the discrete logarithm problem says given a group G and some $g, A \in G$, find $a \in \mathbb{N}$ such that $g^a = A$. The complexity of this problem depends on the group G We extend this idea to the well-known *elliptic curve discrete logarithm problem* (ECDLP) which uses the group structure of points on an elliptic curve as discussed in Part I. We finally discuss Pollard's ρ algorithm which is a highly memory efficient yet greedy algorithm for solving the ECDLP.

Part I
Theoretical

2 Part I: Projective Plane, $\mathbb{P}^2(\mathbb{R})$

A projective plane can be considered as an extension to the Euclidean plane. Suppose we have two lines in the Euclidean plane, then they either intersect at exactly one point or do not intersect, the latter being parallel lines. The projective plane has an additional structure namely, *points at infinity* which allow parallel lines to intersect. It follows any two distinct lines in the projective plane must intersect at exactly one point. We motivate the notion of a point at infinity which is a core object to construct a group structure in the set of points on an elliptic curve.

2.1 Equivalence relation defining $\mathbb{P}^2(\mathbb{R})$

Definition 2.1.1. Let $\sim \subseteq \mathbb{R}^3 \times \mathbb{R}^3$ be an equivalence relation such that for any $\mathbf{x} = (x_1, x_2, x_3)$, $\mathbf{y} = (y_1, y_2, y_3) \in \mathbb{R}^3$, we have that $\mathbf{x} \sim \mathbf{y}$ if and only if $\mathbf{x} = k\mathbf{y}$ for some $k \in \mathbb{R}_{\neq 0}$.

Definition 2.1.2. (Real Projective Plane)
Let $[x] := \{\mathbf{a} \in \mathbb{R}^3 : \mathbf{x} \sim \mathbf{a}\}$ be the equivalence class of $\mathbf{x} \in \mathbb{R}^3$.
The **Real Projective Plane**, $\mathbb{P}^2(\mathbb{R})$, is the collection of all equivalence classes:

$$\mathbb{P}^2(\mathbb{R}) := \bigcup_{x \in \mathbb{R}^3} [\mathbf{x}], \text{ with } [x] = (x_1, x_2, x_3) \text{ such that } x_1, x_2, x_3 \neq 0$$

For a geometric representation, we can construct a map from **lines** in the $\mathbb{R}^2$ that pass through the origin, to **points** in the projective plane, $\mathbb{P}^2(\mathbb{R})$.

Example 2.1. Let $(x_1, x_2) \in \mathbb{R}^2$ with $x_1, x_2 \neq 0$. We construct a unique line passing through $(0, 0)$ and (x_1, x_2):

$$L = \{(kx_1, kx_2) \in \mathbb{R}^2 : k \in \mathbb{R}\} \subseteq \mathbb{R}^2$$

Now, let $f : L \to \mathbb{P}^2(\mathbb{R})$ be a map such that:

$$f\Big((kx_1, kx_2)\Big) := [x_1, x_2, c] \in \mathbb{P}^2(\mathbb{R}), \text{ for some } c \in \mathbb{R}_{\neq 0}$$

Notice the equivalence class:

$$[kx_1, kx_2, kc] = \{(kx_1, kx_2, kc) \in \mathbb{R}^3 : k \in \mathbb{R}\}$$

is a **point** in $\mathbb{P}^2(\mathbb{R})$ and represents *points in* $\mathbb{R}^2$ along the line L.

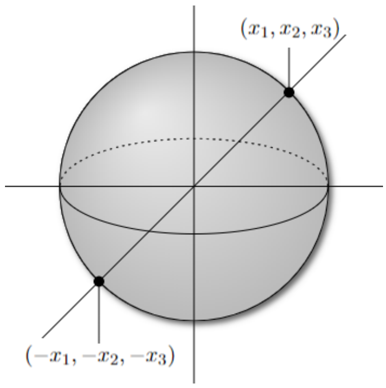

Figure 1: The real projective space, $\mathbb{P}^2(\mathbb{R})$ can be viewed in $\mathbb{R}^2$ as points lying on the upper hemisphere as seen in example 2.1. Image Source: Primary

A natural question that arises is the case of $c = 0$. In this case, we can not apply our current definition to such points. Points with $c = 0$ forms our understanding of **lines** and the notion of **infinity** in the projective plane.

2.2 Points and lines at infinity in $\mathbb{P}^2(\mathbb{R})$

Definition 2.2.1. The line $\langle a_1, a_2, a_3\rangle \subseteq \mathbb{P}^2(\mathbb{R})$ is the collection of points $[\mathbf{x}] = [x_1, x_2, x_3] \in \mathbb{P}^2(\mathbb{R})$ that satisfy:

$$a_1x_1 + a_2x_2 + a_3x_3 = 0, \text{ where } a_1, a_2, a_3 \in \mathbb{R}_{\neq 0}$$

Remark. We use square brackets [] to denote *points* and angled brackets $\langle\rangle$ to denote *lines* in the projective plane $\mathbb{P}^2(\mathbb{R})$.

Before introducing the line at infinity, we describe the points at infinity that lie on this line. For this we suppose $c = 0$ from earlier.

Definition 2.2.2. Let $\mathbf{x} = [x_1, x_2, 0] \in \mathbb{P}^2(\mathbb{R})$ with $x_1, x_2 \neq 0$. Then $\mathbf{x}$ is a **point at infinity**.

Notice how such points, $[x_1, x_2, 0]$ satisfy the following:

$$0x_1 + 0x_2 + 1(0) = 0$$

Such points thus lie on the line $\langle 0, 0, 1\rangle$.

Definition 2.2.3. The **line at infinity**, $\langle 0, 0, 1\rangle$, is the set of points at infinity.

We now prove the following proposition:

Proposition 2.2.4. *Let $\langle a_1, a_2, a_3\rangle$ and $\langle b_1, b_2, b_3\rangle$ be two lines in the real projective plane. Then there exists a unique point $[x_1, x_2, x_3] \in \mathbb{P}^2(\mathbb{R})$ through these two lines.*

Proof. For $x_1, x_2, x_3 \in \mathbb{R}_{\neq 0}$ consider the following equations:

$$a_1x_1 + a_2x_2 + a_3x_3 = 0 \tag{1}$$
$$b_1x_1 + b_2x_2 + b_3x_3 = 0 \tag{2}$$

To show the point $\mathbf{x} = [x_1, x_2, x_3] \in \mathbb{P}^2(\mathbb{R})$ lies on both $\langle a_1, a_2, a_3\rangle$ and $\langle b_1, b_2, b_3\rangle$ we must find unique $x_1, x_2, x_3 \in \mathbb{R}_{\neq 0}$ that solve the system above. Consider the system rewritten in matrix form below:

$$\begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Since the two lines are distinct, it follows the rows of $\begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{pmatrix}$ are linearly independent. Therefore, from *MA106* we understand there must be a unique solution to the system above. $\square$

Corollary 2.2.5. *Let L be a line in the real projective plane. Then L has exactly one point at infinity.*

Proof. By 2.2.4 L and the line at infinity, $\langle 0, 0, 1\rangle$, intersect at a unique point $[x_1, x_2, x_3] \in \mathbb{P}^2(\mathbb{R})$. By definition, $\langle 0, 0, 1\rangle$ is the set of points at infinity, hence it follows $x_3 = 0$ so, $[x_1, x_2, x_3] = [x_1, x_2, 0]$ which is a point at infinity. Therefore L contains exactly one point at infinity. $\square$

2.3 Elliptic curves

We generally define what elliptic curve are and use the theory on the real projective plane to motivate how points on a given elliptic curve form an abelian group.

Definition 2.3.1. An *elliptic curve* over a field K is a general homogeneous cubic curve defined by:

$$y^2z = x^3 + axz^2 + bz^3$$

which can be simplified into *Weierstrass normal form* Silverman and Tate [15, p. 16], by letting $z = 1$:

$$y^2 = x^3 + ax + b$$

with $a, b \in K$ and $\Delta := -16(4a^3 + 27b^3) \neq 0$

Remark. By definition, elliptic curves have $\Delta \neq 0$ to ensure there are no *singular points.* This ensures that every point on the curve has a well-defined tangent line which is essential for section 2.4. See [15, p. 78]

Example 2.2. An elliptic curve generally takes this shape in $K := \mathbb{P}^2(\mathbb{R})$

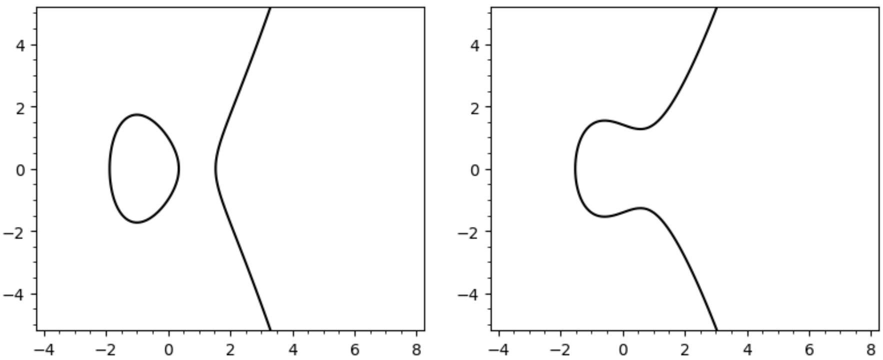

Figure 2: $y^2 = x^3 - 3x + 1$ $(\Delta = 1296)$; $y^2 = x^3 - x + 2$ $(\Delta = -1664)$ respectively, both are elliptic curves. Generated using SAGE.

2.4 Group structure on elliptic curves

We introduce an *addition* binary operation, denoted $\oplus$, that turns the points on an elliptic curve into an abelian group. This section is motivated by Jousma [7].

Definition 2.4.1. We define

$$E_{a,b}(\mathbb{P}^2(\mathbb{R})) := \{[x, y, z] \in \mathbb{P}^2(\mathbb{R}) : y^2z = x^3 + axz^2 + bz^3\} \cup \{\mathcal{O}\} \subseteq \mathbb{P}^2(\mathbb{R})$$

to be the set of points on a general elliptic curve in $\mathbb{P}^2(\mathbb{R})$ with arbitrary $a, b \in \mathbb{R}$ and $\mathcal{O} := [0, 1, 0]$ being the point at infinity from definition 2.2.2.

For a better geometric interpretation, let us consider:

$$E_{a,b}(\mathbb{R}) := \{(x,y) \in \mathbb{R}^2 : y^2 = x^3 + ax + b\} \cup \{\mathcal{O}\}$$

Let $P_1, P_2 \in E_{a,b}(\mathbb{R})$ be two points on some elliptic curve, $E_{a,b}$ and denote the line passing through P_1 and P_2 by L. Since $E_{a,b}$ is *non-singular* i.e $\Delta \neq 0$, L must intersect $E_{a,b}$ in a third point, call it P_3. Reflecting P_3 across the x-axis gives us precisely $P_1 \oplus P_2$.

Assume $P_1 \neq P_2$ with $x_1 \neq x_2$ with $P_1 = (x_1, y_1)$, $P_2 = (x_2, y_2)$ and the line L through P_1 and P_2 by the equation $y = cx + d$. Then $c = \frac{y_1 - y_2}{x_1 - x_2}$, so we can find all intersection of L and $E_{a,b}$ by substituting and getting:

$$(cx + d)^2 = x^3 + ax + b$$

which we can rewrite as

$$x^3 - c^2x^2 + (a - 2cd)x + (b - d^2) = 0$$

The above has three roots in the algebraic closure of $\mathbb{R}$ which is $\mathbb{C}$ (with multiplicity). However, we know x_1 and x_2 are two of the roots, denote the third root by x_3. So we have:

$$x^3 - c^2x^2 + (a - 2cd)x + (b - d^2) = (x - x_1)(x - x_2)(x - x_3) \tag{3}$$

and by expanding and equating we get $x_3 = c^2 - x_1 - x_2$. This is the x-coordinate of $P_1 \oplus P_2$. We can find y_3 by using L and reflecting across the x-axis, so $y_3 = c(x_1 - x_3) - y_1$. Thus $P_1 \oplus P_2 = (x_3, y_3)$.

Now suppose $P_1 \neq P_2$ but $x_1 = x_2$. In this case L is vertical and will intersect at most two points on $E_{a,b}$. In $\mathbb{P}^2(\mathbb{R})$, by corollary 2.2.5, $\mathcal{O} \in L$, reflecting $\mathcal{O}$ across the x-axis gives $\mathcal{O}$ so $P_1 \oplus P_2 = \mathcal{O}$.

Now assume $P_1 = P_2 = (x_1, y_1)$. Let L be the tangent line at P_1. The slope of L, denoted c, is given by differentiating $E_{a,b}$ and substituting P_1:

$$c = \frac{3x_1^2 + a}{2y_1}$$

If $y_1 = 0$ then the tangent line is vertical, so by the previous case $P_1 \oplus P_2 = \mathcal{O}$.
Now we may assume $y_1 \neq 0$, then we proceed similar to the first case. However, x_1 is our only known root of (3). Since this is a root for P_1 and P_2, it is a double root so we can find $x_3 = c^2 - 2x_1$ and $y_3 = c(x_1 - x_3) - y_1$ hence $P_1 \oplus P_2 = (x_3, y_3)$.

Finally, we take the case $P_1 = \mathcal{O}$, then the line L through P_1 and P_2 is vertical. The third point of intersection is the reflection of P_2 across the x-axis. By definition of $\oplus$ we must reflect back across the x-axis which returns P_2 so we have:

$$\mathcal{O} + P_2 = P_2 = P_2 + \mathcal{O} \tag{4}$$

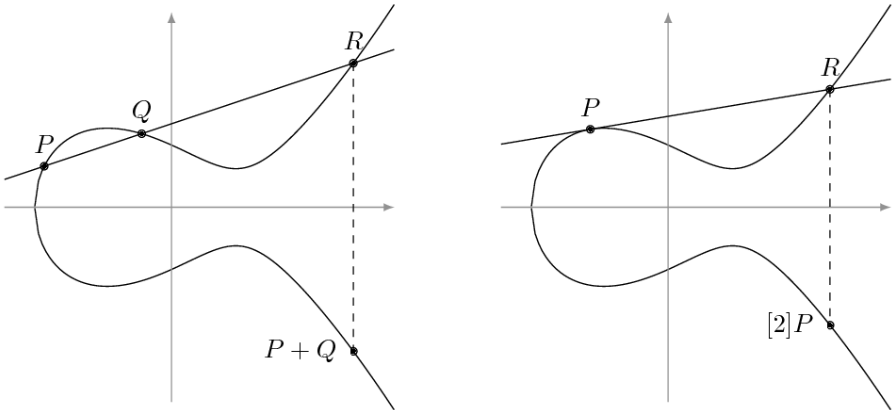

Figure 3: Geometric interpretation of the group law on $E_{a,b}(\mathbb{R})$, from Feo [5]

We can now state the following theorem that turns $E_{a,b}(\mathbb{P}^2(\mathbb{R}))$ into an abelian group. For simplification of notation, we use $E_{a,b}$.

Key Theorem 2.4.2. *The addition law, $\oplus$ as defined above on $E_{a,b}$, has the following properties, for all $P, Q, R \in E_{a,b}$:*

(i) $P \oplus \mathcal{O} = \mathcal{O} \oplus P = P$ *(Existence of identity)*

(ii) $P \oplus (-P) = \mathcal{O}$ *(Existence of inverses)*

(iii) $P \oplus Q = Q \oplus P$ *(Commutativity)*

(iv) $P \oplus (Q \oplus R) = (P \oplus Q) \oplus R$ *(Associativity)*

and thus $(E_{a,b}, \oplus)$ forms an abelian group.

Proof. (**Identity**): From 4 it follows $\mathcal{O}$ acts as an identity in $E_{a,b}$.
(Inverses): Let $P \in E_{a,b}$, then define $-P$ to be the reflection of P across the x-axis. It follows the line, L, intersecting P and $-P$ is vertical and by symmetry of $E_{a,b}$, will intersect at most two points on $E_{a,b}$. Using corollary 2.2.5 we have $\mathcal{O} \in L$, reflecting this third point across the x-axis gives $\mathcal{O}$ so we have $P + (-P) = \mathcal{O}$ so $-P$ is indeed the inverse of P.
(Associativity): We omit the proof for associativity due to its complexity, a geometric argument is provided in appendix B.
(Commutativity): This follows from the uniqueness of a line L through points $P, Q \in E_{a,b}$ □

3 Part I: Finite Fields

The aim of this section is to show all finite fields $\mathbb{F}_{p^q}$ with p prime and $k \in \mathbb{Z}$ are isomorphic up to its size. We can then use this alongside the group structure on $E_{a,b}$ to build the intractible problem present in $E_{a,b}$ over finite fields. As a notational remark, we use 1 to denote the multiplicative identity and 0 to denote the additive identity.

Definition 3.0.1. Let $k \in \mathbb{Z}_{\geq 0}$. A **finite field**, $\mathbb{F}$, of *order* k is a field such that $|\mathbb{F}| = k$.

Definition 3.0.2. Let $\mathbb{F}$ be a finite field. There exists a *finite characteristic* on $\mathbb{F}$ if there exists $n \in \mathbb{Z}_{\geq 0}$ such that:

$$\underbrace{1 + 1 + \cdots + 1}_{\text{n times}} = 0$$

in $\mathbb{F}$. The smallest such n is defined as the **characteristic** of $\mathbb{F}$.

Proposition 3.0.3. *Every finite field has finite characteristic p, with p prime.*

Proof. Let $\mathbb{F}$ be a finite field of order k. Then $\mathbb{F} = \{1, 2, \ldots, k\}$ for some $k \in \mathbb{Z}_{\geq 0}$. Let $m, n \in \mathbb{Z}_{\geq 0}$ be such that:

$$m \equiv n \pmod{k} \iff m - n \equiv 0 \pmod{k}$$

It directly follows:

$$n - m = \underbrace{1 + 1 + \cdots + 1}_{\text{n - m times}} = 0 \text{ in } \mathbb{F}$$

Suppose, for a contradiction, the characteristic of $\mathbb{F}$ is k, with k composite. Then for some $1 < p, q < k$ we have:

$$\underbrace{(1 + 1 \cdots + 1)}_{\text{p times}} \cdot \underbrace{(1 + 1 \cdots + 1)}_{\text{q times}} = \underbrace{1 + 1 \cdots + 1}_{\text{k times}} = 0$$

Thus $pq = 0$ so $pq \neq 1$ which contradicts p having a multiplicative inverse. Thus k must be prime. ☐

Example 3.1. The most common examples of a finite field are the integers modulo p, $\mathbb{Z}/p\mathbb{Z}$, with p prime. Recall $\mathbb{Z}/p\mathbb{Z} = \{0, 1, ..., p - 1\}$.

3.1 Classification of finite fields

We use the polynomial $x^{p^d} - x$ to classify finite fields.

Definition 3.1.1. Let $\mathbb{F}$ be a field. A subset, K, of $\mathbb{F}$ that is also a field under the operations of $\mathbb{F}$ is called a **subfield** of $\mathbb{F}$.

Definition 3.1.2. A field containing no proper subfields is called a **prime** field.

Definition 3.1.3. The intersection of all subfields of a field, $\mathbb{F}$, is itself a field called the **prime subfield** of $\mathbb{F}$. We denote this as $P(\mathbb{F})$.

Definition 3.1.4. Let $\mathbb{F}$ and K be fields. Suppose $f(x) \in K[x]$ factors in $\mathbb{F}[x]$, we then say f *splits* in $\mathbb{F}$. Moreover, if $\mathbb{F}$ is the smallest such field we then say $\mathbb{F}$ is the *splitting field* of f over K.

These definitions are taken from Neunhöffer [13, Ch. 2, p. 1], and will help build sufficient theory to characterise finite fields up to their size.

Remark. In the following lemmas and propositions, we denote $\mathbb{Z}\backslash p\mathbb{Z}$ as $\mathbb{F}_p$ to simplify notation.

Lemma 3.1.5. *The prime subfield of a field, $\mathbb{F}$, is isomorphic to $\mathbb{F}_p$ if $\mathbb{F}$ has characteristic p.*

Proof. Any subfield of $\mathbb{F}$ must contain the additive and multiplicative identities, 0 and 1, so by closure and having characteristic p, it must contain the following set:

$$Q(\mathbb{F}) := \{0 \cdot (1_{\mathbb{F}}), 1 \cdot (1_{\mathbb{F}}), 2 \cdot (1_{\mathbb{F}}), \cdots, (p-1) \cdot (1_{\mathbb{F}})\}$$

Since every subfield contains $Q(\mathbb{F})$ it follows from the definition of $P(\mathbb{F})$:

$$Q(\mathbb{F}) \subseteq P(\mathbb{F})$$

We can also easily verify that $Q(\mathbb{F})$ is a subfield of $\mathbb{F}$ therefore $P(\mathbb{F})$, which is the intersection of all subfields of $\mathbb{F}$, must be contained in $Q(\mathbb{F})$ i.e:

$$P(\mathbb{F}) \subseteq Q(\mathbb{F})$$

Therefore, $Q(\mathbb{F}) = P(\mathbb{F})$ and by construction, $Q(\mathbb{F}) \cong \mathbb{F}_p$ so it follows $P(\mathbb{F}) \cong \mathbb{F}_p$. $\square$

Lemma 3.1.6. *Let $\mathbb{F}$ be a finite field containing a subfield K with q elements. Then F has q^m elements, where $m := [\mathbb{F} : K]$.*

Proof. Since $\mathbb{F}$ forms a vector space over any of its subfields, we have specifically $\mathbb{F}$ is a vector space over K and is finite dimensional since $\mathbb{F}$ is finite.
Let $m := [\mathbb{F} : K]$ be the dimension of $\mathbb{F}$ over K, then $\mathbb{F}$ has a basis with respect to K of m elements $\{b_1, ..., b_m\} \subseteq \mathbb{F}$.
Therefore, each element of $\mathbb{F}$ can be expressed as:

$$k_1 b_1 + \cdots + k_m b_m \text{ with } k_1, ..., k_m \in K$$

Since each k_i can take up to $|K| = q$ choices, we have $|\mathbb{F}| = q^m$ $\square$

Using the above two lemmas we can prove the following:

Theorem 3.1.7 (Classification of finite fields)**.** *Let $\mathbb{F}$ be a finite field. Then $\mathbb{F}$ has p^n elements, where the prime, p, is the characteristic of $\mathbb{F}$ and n is the dimension of $\mathbb{F}$ over its prime subfield.*

Proof. Since $\mathbb{F}$ is finite, by proposition 3.0.3, it must have characteristic p, with p prime. Then by lemma 3.1.5, the prime subfield, $P(\mathbb{F}) \cong \mathbb{F}_p$ and $|F_p| = p$ so by applying lemma 3.1.6, $|\mathbb{F}| = p^n$ where n is the dimension of the vector space formed by $\mathbb{F}$ over its prime subfield, $P(\mathbb{F})$. $\square$

We have shown that all finite fields must have prime power order so there cannot exist any finite fields of order 6. Our next goal is to show, given a prime power, p^n, there exists a finite field with p^n elements.

Lemma 3.1.8 (Fermat's little theorem for finite fields)**.** *Suppose $\mathbb{F}$ is a finite field with q elements, then $\forall a \in \mathbb{F}$ we have:*

$$a^q \equiv a \pmod q \iff a^{q-1} \equiv 1 \pmod q$$

or equivalently, $a^q = a$ in $\mathbb{F}$.

Proof. Let $a \equiv 0 \pmod q$, then it follows $a^q \equiv 0 \equiv a \pmod q$. Now consider the set:

$$\mathbb{F}^* := \{a \in \mathbb{F} : a \neq 0\}$$

this forms a group under multiplication modulo m as seen in *MA136*. By Lagrange's theorem, we have $a^{|\mathbb{F}^*|} = 1_{\mathbb{F}}$. It follows that:

$$a^{|\mathbb{F}^*|} = a^{q-1} = 1_{\mathbb{F}^*}$$

so,

$$a^{q-1} \equiv 1 \pmod q \iff a^q \equiv a \pmod q \quad \text{for } a \in \mathbb{F}^*$$

This completes the proof. □

Lemma 3.1.9. *Suppose $\mathbb{F}$ is a finite field with q elements and K is a subfield of $\mathbb{F}$, then the polynomial $x^q - x$ in $K[x]$ factors in $\mathbb{F}[x]$ as:*

$$x^q - x = \prod_{a \in \mathbb{F}} (x - a)$$

moreover, $\mathbb{F}$ is the splitting field of $x^q - x$ over K.

Proof. Since the polynomial $x^q - x$ has degree q, by the general roots lemma in *MA257* (see appendix C) it has **at most** q roots in $\mathbb{F}$.
By lemma 3.1.8, we have $\forall a \in \mathbb{F}$, $a^q = a$ in $\mathbb{F}$ therefore, $a^q - a = 0$ in $\mathbb{F}$. So all elements of $\mathbb{F}$ are roots of $x^q - x$ therefore, the polynomial splits over $\mathbb{F}$.
Since all elements of $\mathbb{F}$ are roots of $x^q - x$, there cannot exist any smaller field in which $x^q - x$ splits, therefore by definition $\mathbb{F}$ is the splitting field of $x^q - x$ over K. □

Lemma 3.1.10. *Let $f \in \mathbb{F}[x]$ be a nonconstant polynomial. Then $\alpha \in \mathbb{F}$ is a multiple root of f if and only if $f(\alpha) = Df(\alpha) = 0$, where Df is the derivative of f.*

Proof. Let $f(x) = (x - \alpha)^m g(x)$ with m equal to the multiplicity of α in f and $g \in \mathbb{F}$ such that $g(\alpha) \neq 0$. If $m = 0$, then $f(\alpha) = g(\alpha) \neq 0$.
Otherwise,

$$Df = m(x - \alpha)^{m-1} g(x) + (x - a)^m Dg(x)$$

If $m = 1$, then $Df(\alpha) = g(\alpha) \neq 0$. If $m \geq 2$, then $f(\alpha) = Df(\alpha) = 0$.
Thus we have that $\alpha \in \mathbb{F}$ is a multiple root of $f \iff m \geq 2 \iff f(\alpha) = Df(\alpha) = 0$. □

Lemma 3.1.11. *Suppose $\mathbb{F}$ is a field with p^n elements and $a, b \in \mathbb{F}$. Then:*

$$(a+b)^n = a^n + b^n \text{ in } \mathbb{F}$$

Proof. Using lemma 3.1.8 we have $k^q = k$ in $\mathbb{F}$, for all $k \in \mathbb{F}$ therefore letting $k = a+b$ we have $(a+b)^q = a+b$ then applying Fermat's little theorem again gives us $a = a^q$ and $b = b^q$ so we have $(a+b)^q = a^q + b^q$ in $\mathbb{F}$ □

Key Theorem 3.1.12 (Existence and uniquness of finite fields)**.** *Let p be a prime and $n \in \mathbb{N}$. Then there exists a finite field with p^n elements.*
Any finite field with p^n elements is isomorphic to the splitting field of $x^{p^n} - x$ over $\mathbb{F}_p$

Proof. (*Existence*) Consider the polynomial $q(x) = x^{p^n} - x$ over $\mathbb{F}_p$ and let K be its splitting field. We observe:

$$Dq(x) = p^n x^{p^n - 1} - 1 \equiv -1 \pmod p$$

Therefore, by the contrapositive of lemma 3.1.10, q cannot have any multiple roots. So q has exactly p^n distinct roots in K.
Observe that $0^{p^n} - 0 \equiv 0 \pmod p$ and $1^{p^n} - 1 \equiv 0 \pmod p$ so $0, 1 \in K$. Define the set of roots of q as:

$$S := \{\alpha \in K : \alpha^{p^n} - \alpha \equiv 0_K \pmod p\}$$

Suppose $r, s \in S$, then $r^{p^n} \equiv r \pmod p$ and $s^{p^n} \equiv s \pmod p$ so:

$$(rs)^{p^n} \equiv r^{p^n} s^{p^n} \equiv rs \pmod p \iff (rs)^{p^n} - rs \equiv 0 \pmod p$$

so $rs \in S$. Using lemma 3.1.11, we also have:

$$(r-s)^{p^n} \equiv r^{p^n} - s^{p^n} \equiv r - s \pmod p \iff (r-s)^{p^n} - (r-s) \equiv 0 \pmod p$$

so $r - s \in S$. Finally, suppose $r \neq 0$ then:

$$(sr^{-1})^{p^n} \equiv s^{p^n} r^{-p^n} \equiv sr^{-1} \pmod p \iff (sr^{-1})^{p^n} - (sr^{-1}) \equiv 0 \pmod p$$

so $sr^{-1} \in S$. By the subfield criterion (appendix A), S is a subfield of K. But since q splits in K, we have $S = K$. This proves existence of a finite field with p^n elements.
(*Uniqueness*) Suppose K is a field with p^n elements, then by the same argument above we can show that $x^{p^n} - x$ has p^n distinct roots in K, so K is a splitting field for $x^{p^n} - x$. Uniqueness (up to isomorphism) of K follows from the uniqueness of splitting fields from Galois theory as seen in Leinster [10, p. 91]. □

From theorem 3.1.7 and key theorem 3.1.12, we have shown that every finite field is isomorphic up to its size, specifically every finite field, $\mathbb{F}$, has size p^n where p is its prime characteristic and n is the dimension of $\mathbb{F}$ over its prime subfield. Moreover, for any prime p and integer $n \geq 1$, we can finite field with p^n elements. We can now speak of **the** finite field of p^n elements.

3.2 Elliptic curves over a finite field

We can reduce points on an elliptic curve modulo some prime p to get the finite set, $E_{a,b,p}$, described in section 4.2. Thus every element in $E_{a,b,p}$ is an element of F_p which is **the** finite field of p elements. It uses the same group operation as detailed in section 2.4. This finite group is the basis of ECC and will be used in the general *discrete logarithm problem* described in section 4.

Example 3.2. These are the elliptic curves in example 2.1 over the finite field $\mathbb{F}_{101}$

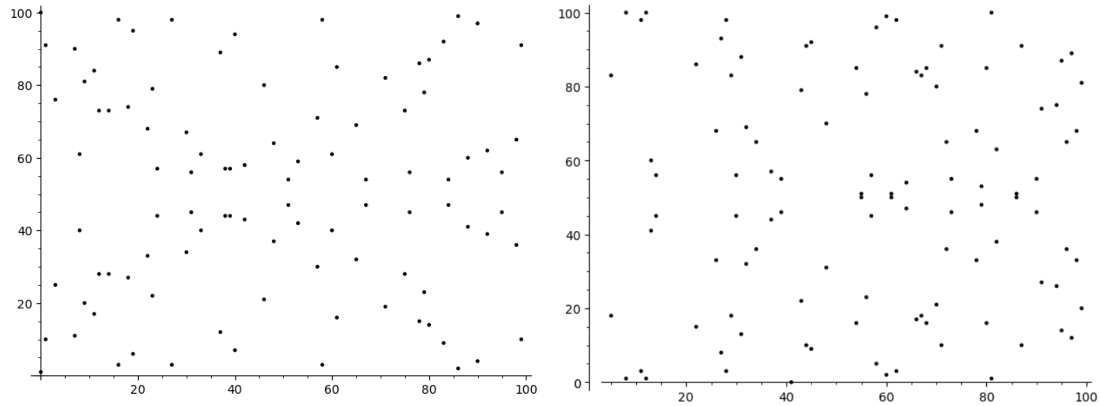

Figure 4: (Left) $y^2 = x^3 - 3x + 1 \pmod{101}$ and (Right) $y^2 = x^3 - x + 2 \pmod{101}$ Generated using SAGE.

Part II
Application

4 Part II: Discrete Logarithm Problem

Before introducing Elliptic Curve Cryptography, we introduce the notion of a cryptosystem to motivate the discrete logarithm problem. A cryptosystem allows parties, *Alice* and *Bob*, to communicate and exchange information over an **insecure** channel such that any adversary, *Eve*, is unable to determine the complete information being sent. A cryptosystem uses a *hard to reverse* task to make reversing the computational processes difficult, the discrete logarithm problem is a core example of a *hard to reverse* task.

4.1 Cryptosystems and their dependence on problems in NP

We list examples of cryptosystems and the *hard* task that allows these systems to operate.

Definition 4.1.1. A decision problem can be solved in *non-deterministic polynomial time* (is in **NP**) if *"yes"* answers can be certified in polynomial time.

Similarly, a decision problem can be solved in *polynomial time* (is in **P**) if the running time on the input n is $\mathcal{O}(n^k)$ for some positive integer k.

In other words, a problem that is in **NP** is hard to solve directly but a solution can be easily verified in polynomial time. Cryptosystems rely on this structure since it means[1] there is no efficient algorithm to **directly** solve the problem in polynomial time. If there existed an efficient algorithm then this problem would be in $\mathbf{P} \cap \mathbf{NP}$ which would mean it could be solved easily and thus rendered useless for a cryptosystem.
By [1] the problems used in the following cryptosystems are assumed to be in **NP**:

Algorithm 4.1. (RSA): Milanov [11, p. 8] The following demonstrates how an encryption cryptosystem works:

Steps :	**Worked Example** :
Step 1: Choose p, q prime	Let $p = 37$, $q = 43$
Step 2: Compute $N := pq$	$N = 1591$
Step 3: Compute $\phi(N)^2 = (p-1)(q-1)$	$\phi(33) = 36(42) = 1512$
Step 4: Choose e such that $1 < e < \phi(N)$ and $\gcd(e, \phi(N)) = 1$	Let $e = 575$
Step 5: Compute d such that $de \equiv 1 \pmod{\phi(N)}$	Let $d = 71$

The *public key* is $(e, N) \leftrightarrow (575, 1591)$ and our *private key* is $(d, N) \leftrightarrow (71, 1591)$. Notice the *secret* part is d which requires $\phi(N)$ to compute. But $\phi(N)$ requires p, q to compute. It is a considerably *hard* task to determine which two primes product to give N, this computational problem is known as the *integer factorisation problem* A Menezes [1, c. 8.2].
Let $m = 1415$ represent our message[3] encoded in integer form. We **encrypt** as follows $m_E := m^e \pmod{N}$ so $m_E = 1415^{575} \equiv 1247 \pmod{1512}$.
Then $m_E = 1247$ is our encrypted message and to **decrypt** we do a similar process using the secret key $m_D = m_E^d \pmod{N}$ so $m_D = 1247^{71} \equiv 1415 \pmod{1512}$ and we have $m = 1415$ back again.

[1]It is an open problem whether $\mathbf{P} = \mathbf{NP}$. Therefore, there may exist (although unlikely) efficient algorithms that break cryptosystems (if $\mathbf{P} = \mathbf{NP}$).

[2]ϕ is the Euler Totient function, we assume without proof a standard property namely: $\phi(pq) = \phi(p)\phi(q)$ with p, q prime

[3]The algorithm requires $1 < m < N$ to avoid *collisions* so any message with encoded length greater than N is broken down into *encoded packets* of size at most $N - 1$.

Algorithm 4.2. (Diffe-Hellman): Silverman [14, p. 4] The following demonstrates how a key-exchange cryptosystem works:
Let G be a group and let $g \in G$ with order n. Take $G = \mathbb{F}_{179}^*$ and consider $150 \in \mathbb{F}_{179}^*$ with order 37.

Steps :	**Worked Example** :
Step 1: *Alice* chooses a secret key $0 < a < n$	Let $a = 17$
Step 2: *Bob* chooses a secret key $0 < b < n$	Let $b = 22$
Step 3: *Alice* computes $A := g^a$	$A = 150^{17} \equiv 92 \pmod{179}$
Step 4: *Bob* computes $B := g^b$	$B = 150^{22} \equiv 168 \pmod{179}$
Step 5: *Alice* and *Bob* exchange A and B	
Step 6: *Alice* computes $B^a = g^{ba}$	$168^{17} \equiv 17 \pmod{179}$
Step 7: *Bob* computes $A^b = g^{ab}$	$92^{22} \equiv 17 \pmod{179}$

17 is the *secret exchanged key*. Notice the only information communicated publicly is $A = g^a$ and $B = g^b$. But one of either a or b is required to compute the *secret exchanged key*. Without loss of generality, to determine a from $g^a = A$ is the *hard to reverse* problem in this cryptosystem. This computational problem is known as the *discrete logarithm problem*

4.2 Elliptic curve discrete logarithm problem, ECDLP

Let p be a prime number and $q = 1$ then by key theorem 3.1.12 consider **the** unique (up to size) finite field:

$$\mathbb{F}_p = \{0, 1, 2, ..., p - 1\}$$

We then similar to definition 2.4.1, set

$$E_{a,b,p} := \{[x, y, 1] \in \mathbb{P}^2(\mathbb{R}) : y^2 \equiv x^3 + ax + b \pmod{p}\} \cup \{\mathcal{O}\}$$

for some $a, b \in \mathbb{R}$ which we choose later. Then $E_{a,b,p}$ is a finite set and we have proved it forms a group in key theorem 2.4.2. Assume $P, Q \in E_{a,b,p}$ satisfy:

$$Q = nP = \underbrace{P \oplus \cdots \oplus P}_{\text{n times}}$$

where $\oplus$ is simply the addition operation discussed in section 2.4. Then to determine n is known as the elliptic curve discrete logarithm problem (**ECDLP**). Notice its similarity to the general discrete logarithm problem in algorithm 4.2. **ECDLP** is a *hard*[4] problem assumed to be in **NP**

[4]The hardness of this problem also depends on the elliptic curve used in the group. There is no known proof to determine which elliptic curves are "secure" thus we rely on standardised curves Bernstein and Lange [2]

5 Part II: Elliptic Curve Cryptography

5.1 Pollard's ρ algorithm for solving ECDLP

Let $G := E_{a,b,p}$ be the set of points on some elliptic curve, Pollard's ρ algorithm is a probabalistic[5] algorithm for computing the discrete logarithm, we specifically look at the ECDLP.
It generates a pseudo-random sequence of points in $E_{a,b,p}$ using an *iterating* function $x_{i+1} = f(x_i)$ with $x_i \in G$. Since we define G to be over a finite field, G is a finite group, therefore the sequence $\{x_i\} \subset E_{a,b,p}$ is finite and thus will meet an element that has occured before. This is known as a collision which we find using *Floyd's cycle-finding algorithm*[6]
We define a pseudo-random function similar to that used by Pollard, let $P, Q \in G$, define $f : G \to G$ by:

$$f(X) = \begin{cases} P \oplus X & X_x \equiv 0 \pmod 3 \\ X^2 = X \oplus X & X_x \equiv 1 \pmod 3 \\ Q \oplus X & X_x \equiv 2 \pmod 3 \end{cases}$$

where X_x is the x-coordinate of X. The ECDLP appears hard to solve due to the randomness when taking powers of some element in G. We exploit this to introduce the pseudo-randomness in f since we cannot directly determine what $X_x \pmod 3$ is.
We further define maps $g : G \times \mathbb{Z} \to \mathbb{Z}$ and $h : G \times \mathbb{Z} \to \mathbb{Z}$ by:

$$g(X, k) = \begin{cases} k & X_x \equiv 0 \pmod 3 \\ 2k \pmod n & X_x \equiv 1 \pmod 3 \\ k + 1 \pmod n & X_x \equiv 2 \pmod 3 \end{cases}$$

and

$$h(X, k) = \begin{cases} k + 1 \pmod n & X_x \equiv 0 \pmod 3 \\ 2k \pmod n & X_x \equiv 1 \pmod 3 \\ k & X_x \equiv 2 \pmod 3 \end{cases}$$

these are also pseudo-random maps. Using the Pollard ρ algorithm outlined in appendix E, we are returned with an array $[a, b, A, B]$ or *failure*. Assuming the algorithm is probabalistically successful, we can use this array to solve the ECDLP. From the algorithm, $[a, b, A, B]$ satisfy the following relation:

$$aP + bQ = AP + BQ$$

[5]A probabalistic (randomised) algorithm employs some degree of randomness into its procedure

[6]Given a finite set S and a function $F : S \to S$, we use a starting point $x_0 \in S$ and define $x_{i+1} = F(x_i)$ to generate a sequence X. We want to find a collision i.e $X_i = X_j$. It is important to assume F behaves randomly. Floyd defines a new sequence $Y_0 = X_0$ and $Y_{i+1} = F(F(Y_i))$ to then find the first index, t such that $Y_t = X_t$. Joux [8, p. 225]

where $P, Q \in E_{a,b,p}$. We can determine n as described in section 4.2 using the extended Euclidean algorithm.

$$\implies \quad (a - A)P = (B - b)Q = (B - b)nP$$

So we can reduce the ECDLP to the following linear congruence:

$$(a - A) \equiv (B - b)n \pmod{p}$$

which can be solved if and only if $\gcd(B - b, p) \mid (a - A)$ as seen in *MA257*.

6 Conclusion

We have used projective geometry to motivate the notion of a *point at infinity* which subsequently led to the underlying group structure present in the set of points on an elliptic curve. We discretised this abelian group over a finite field which we classified up to its size. Using the discrete logarithm problem and our finite abelian group, we introduced the ECDLP which was the assumed **NP** problem that elliptic curve cryptography relied on. Having set the theory we further discussed a probabalistic algorithm for solving the ECDLP namely, Pollard's ρ algorithm. With the openness of the **P** = **NP** problem, it is still an open question as to whether there exists a fast, deterministic algorithm for solving the ECDLP on any elliptic curve.

This open area of research extends into the field of quantum computing where algorithms, such as *Shor's algorithm*, aim to solve the same types problems as we have seen in this essay, in polynomial time for which no polynomial time classical algorithms are known. This change in approach allows the potential for introducing polynomial time algorithms which could solve problems in **NP** that seem unsolvable otherwise. This would theoretically mean if quantum computing scaled into a feasible option then our encryption algorithms described in this essay would be rendered useless however, quantum computing is currently an unfeasible practical tool.

For further reading on Peter Shor's quantum algorithm for solving the discrete logarithm problem (DLP), this paper by John Proos [6] is highly recommended.

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Appendices

A Proof of Subfield Criterion

Definition A.1. A **field** is any set F having two (closed) binary operations $+$ and $\cdot$ such that $[F1] - [F9]$ are satisfied:
[**F1**]: The operation $+$ is associative: $a + (b + c) = (a + b) + c$ for all $a, b, c \in F$
[**F2**]: The operation $+$ is commutative: $a + b = b + a$ for all $a, b \in F$
[**F3**]: There is an additive identity $0 \in F$ satisfying $a + 0 = a$ for all $a \in F$
[**F4**]: Every element $a \in F$ has an additive inverse, $-a \in F$, such that $a + (-a) = 0$
[**F5**]: The operation $\cdot$ is associative: $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for all $a, b, c \in F$
[**F6**]: The operation $\cdot$ is commutative: $a \cdot b = b \cdot a$ for all $a, b \in F$
[**F7**]: There is a multiplicative identity $1 \neq 0$, such that $1 \cdot a = a = a \cdot 1$ for all $a \in F$
[**F8**]: Every nonzero $a \in F$ has a multiplicative inverse, $a^{-1} \in F$ such that $a \cdot a^{-1} = 1$
[**F9**]: The operation $\cdot$ distributes over $+$: $a \cdot (b + c) = a \cdot b + a \cdot c$ for all $a, b, c \in F$

Theorem A.2. ***(Subfield Criterion)*** *A subset S of a field $\mathbb{F}$ is a* ***subfield*** *if and only if S contains $0_{\mathbb{F}}$ and $1_{\mathbb{F}}$, and is closed under subtraction and division. In other words, for any $a, b, c \in S$ with $c \neq 0$ we have:*

$$0_{\mathbb{F}}, 1_{\mathbb{F}} \in S, \quad a - b \in S \quad \textit{and} \quad a \cdot c^{-1} \in S$$

The following standard proof is taken from Dummit [4, p. 9]

Proof. Let $\mathbb{F}$ be a field and $S \subseteq \mathbb{F}$.
($\Longrightarrow$) Suppose S is a subfield of $\mathbb{F}$, then by definition it contains the additive identity 0_S.
$0_S + 0_S = 0_S$ by [**F3**] in S and $0_S + 0_{\mathbb{F}} = 0_S$ by [**F3**] in $\mathbb{F}$. So by additive cancellation in $\mathbb{F}$ we have $0_S = 0_{\mathbb{F}}$. Similarly, we have $1_S = 1_{\mathbb{F}}$. The additive and multiplicative inverses in S are the same of those in $\mathbb{F}$ since $S \subseteq \mathbb{F}$ therefore, $a - b \in S$ amd $a \cdot c^{-1} \in S$ for any $a, b, c \in S$.
($\Longleftarrow$) Conversely, suppose S is non-empty ($0_{\mathbb{F}}, 1_{\mathbb{F}} \in S$) and S is closed under subtraction and division. Axioms about associativity and commutativity: [**F1**], [**F2**], [**F5**], [**F6**], [**F9**] follow immediately from $\mathbb{F}$.
Letting $b = a$ gives us $a - a = 0_{\mathbb{F}} \in S$ which is [**F3**] and then letting $a = 0$ gives us $0 - b = -b \in S$ which gives us [**F4**].
Similarly, letting $c = a$ gives us $a \cdot a^{-1} = 1_{\mathbb{F}} \in S$ giving us [**F7**] and finally, letting $a = 1$ gives $c^{-1} \in S$ giving us [**F8**]. Therefore, S is a field, more precisely a subfield of $\mathbb{F}$. $\square$

B Geometric *proof* of associativity of *addition* operation

The associativity law is immensely difficult to prove using undergraduate knowledge. Instead we provide geometric argument which demonstrates why the associativity law holds in theorem 2.4.2. This proof is taken from K. Fujii [9] who also provides a purely algebraic proof using calculations computed by MATHEMATICA.
Let P, Q, R be three points on an elliptic curve then:

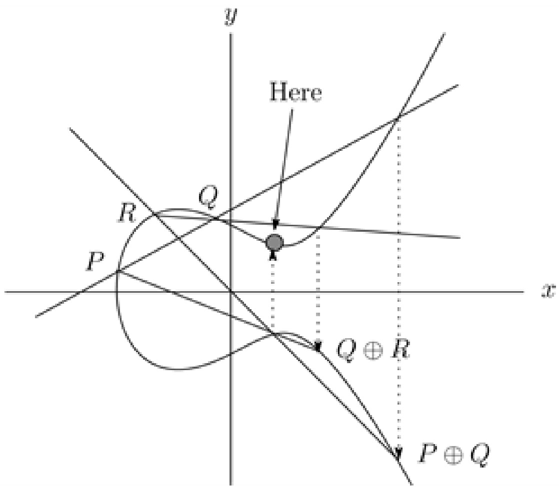

Figure 5: Associativity $(P \oplus Q) \oplus R = P \oplus (Q \oplus R)$

C General Roots Lemma

The following lemma is used in lemma 3.1.9 and the proof structure is taken from Chow [3, p. 15] lecture notes.

Lemma C.1. *Let p be prime, and let $f(x) \in \mathbb{F}_p[x]$ be a non-zero polynomial of degree d. Then f has at most d roots in $\mathbb{F}_p$.*

Proof. This is clear for $d = 0$, so lets assume that $f(\alpha) = 0$ for some $\alpha \in \mathbb{F}_p$ and induct on the degree. By polynomial long division, we can find $q(x) \in \mathbb{F}_p[x]$ and $r \in \mathbb{F}_p$ such that:

$$f(x) = (x - \alpha)q(x) + r, \qquad \deg(q) \leq d - 1$$

Substituting $x = \alpha$ into the above yields $r = 0$, so $f(x) = (x - \alpha)q(x)$. If $\beta \neq \alpha$ is a root of f, then it is a root of q. By our inductive hypothesis, the polynomial q has at most $d - 1$ roots in $\mathbb{F}_p$, so f must have at most d roots. $\square$

D Python package for arithmetic on elliptic curves

Source: Nakov [12] We use a small python package called *tinyec* (https://github.com/alexmgr/tinyec) which enables us to perform arithmetic operations on elliptic curves to reduce tedious algebraic manipulations defined in section 2.4.

```
from tinyec.ec import SubGroup, Curve
from tinyec import registry

# Example of playground elliptic curve
# Defines a finite field
field = SubGroup(p=17, g=(15, 13), n=18, h=1)
# Defines an elliptic curve over the field
curve = Curve(a=0, b=7, field=field, name='p1707')
print('curve:', curve)

for k in range(0, 25):
    p = k * curve.g # Computes iterative powers of the generator of
                                                          curve
    print(f"{k} * G = ({p.x}, {p.y})")
```

E Implementing Pollard's ρ algorithm

Algorithm 1 Pollard's rho algorithm

procedure POLLARDSOLVELOG(P, Q) $\triangleright$ Input two points on $E_{a,b,p}$
  $a_0 \leftarrow 0$, $b_0 \leftarrow 0$, $x_0 \leftarrow P$
  $i \leftarrow 1$
  repeat
    $x_i \leftarrow f(x_{i-1})$,
    $a_i \leftarrow g(x_{i-1}, a_{i-1})$,
    $b_i \leftarrow h(x_{i-1}, b_{i-1})$
    $x_{2i} \leftarrow f(f(x_{2i-2}))$,
    $a_{2i} \leftarrow g(f(x_{2i-2}), g(x_{2i-2}, a_{2i-2}))$,
    $b_{2i} \leftarrow h(f(x_{2i-2}), h(x_{2i-2}, b_{2i-2}))$
    if $x_i = x_{2i}$ **then**
      if $b_i - b_{2i} = 0$ **then**
        return failure
      else
        $x \leftarrow r^{-1}(a_{2i} - a_i) \mod \text{n}$
        return a_i, a_{2i}, b_i, b_{2i}
      end if
    else
      $i \leftarrow i + 1$
    end if
  until return is called
end procedure

We implement the above algorithm into python as follows: Refer to appendix D for the *tinyec* package.

```python
from tinyec.ec import SubGroup, Curve
from tinyec import registry
import time

# Example of playground elliptic curve
field = SubGroup(p=17, g=(15, 13), n=18, h=1)
curve = Curve(a=0, b=7, field=field, name='p1707')
# Defines the curve y^2 = x^3 + 7 modulo 17

print('Curve:', curve)
print('-'*50)
k = 5       # Solution to ECDLP to find

P = curve.g # Generator
Q = k * P   # Element of E_{a, b, p}
n = 18      # Order of the generator P
```

```python
def new_xab(x, a, b):
    ''' Computes x_i, a_i, b_i '''
    if (x.x == None):
        x = a*P + (b + 1)* Q
        b = (b + 1) % n
    elif (x.x % 3 == 0): # in S_1
        x = a*P + (b + 1)* Q
        b = (b + 1) % n
    elif (x.x % 3 == 1): # in S_2
        x = (2*a)*P + (2*b)*Q
        a = (2 * a) % n
        b = (2 * b) % n
    elif (x.x % 3 == 2): # in S_3
        x = (a + 1)*P + b*Q
        a = (a + 1) % n
    return x, a, b

def main():
    ''' Follows the pseudo-code '''
    x = 1 * P; a = 0; b = 0
    X = 2 * P; A = a; B = b
    for i in range(0, n + 1):
        x, a, b = new_xab(x, a, b)
        X1, A1, B1 = new_xab(X, A, B)
        X, A, B = new_xab(X1, A1, B1)
        print("i =", i, "| x = (" + str(x.x) + ", " + str(x.y) + ") [a =", a, ", b = " + str(b) + "] | X = (" + str(X.x) + ", " + str(X.y) + ") [A =", A, ", B = " + str(B) + "]")
        if (x == X):
            ''' Checks the pairs (a, b) and (A, B) give the same point '''
            print('-'*50 + "\nCollision found [a = " + str(a) + ", b = " + str(b) + ", A = " + str(A) + ", B = " + str(B) + "]")
            print("For (a = " + str(a) + ", b = " + str(b) + "), aP + bQ = " + str(a * P + b * Q))
            print("For (A = " + str(A) + ", B = " + str(B) + "), AP + BQ = " + str(A * P + B * Q))
            break

main()
```